

CPEN 416 / EECE 5710

Lecture 21: Stabilizer codes

Tuesday 25 November 2025

Announcements

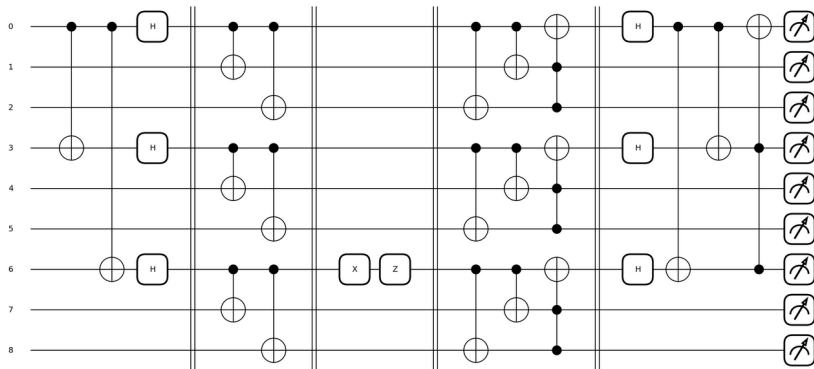
- Quiz 9 today
- TA4 due Friday 23:59
- One more content lecture (Thursday), then presentations

Last time

We made a code to fix phase flip errors by making them look like bit flip errors in a different basis.

Last time

We *concatenated* the bit and phase flip codes to create a 9-qubit code that corrects *any* single-qubit error. But, why does this work?



Learning outcomes

Today:

- ➊ Outline the conditions under which errors can be corrected
- ➋ Define the stabilizers of a quantum error correcting code
- ➌ Express the bit flip, phase flip, Shor code, and 5-qubit code in the stabilizer formalism

Correcting arbitrary errors

The 9-qubit code can correct *arbitrary* errors on a *single qubit*.

Represent possible errors as $\{E_i\}$, full error channel as

We can correct $\{E_i\} = \{I, X_j, Z_j, X_j Z_j\}_j$. If we can write an arbitrary error $\mathcal{E}_k$ on qubit j as

then we can correct it too.

Conditions for quantum error correction

How do we know if a particular error is correctable by a QECC?

Formal definition of a quantum error correcting code is a *subspace*, C , called the **codespace**.

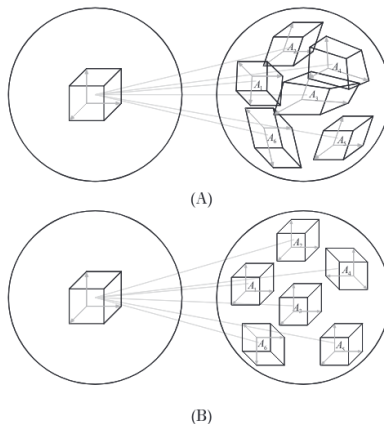
Example: bit flip code.

- Codewords:
- Codespace: contents have the form

Define a **projector** onto the codespace,

Conditions for quantum error correction

The errors this code corrects are all mapped to different **orthogonal** subspaces of the full Hilbert space.



Conditions for quantum error correction

Theorem 10.1: (Quantum error-correction conditions) Let C be a quantum code, and let P be the projector onto C . Suppose $\mathcal{E}$ is a quantum operation with operation elements $\{E_i\}$. A necessary and sufficient condition for the existence of an error-correction operation $\mathcal{R}$ correcting $\mathcal{E}$ on C is that

(10.16)

for some Hermitian matrix α of complex numbers.

If such an $\mathcal{R}$ exists, $\{E_i\}$ is called a *correctable set of errors*.

Correctable errors: bit flip code

Consider $\{I, E_1 = XII, E_2 = IXI, E_3 = IIX\}$

Try including a 2-qubit error:

Example: amplitude damping

Not immediately clear how to express this in terms of no errors, bit, phase or bit/phase flip errors. Can we still correct it?

Straightforward to re-express K_1 :

For K_0 ,

Bit flip code: recovery revisited

Last week / last class, we considered two recovery circuits for the bit flip code.

Stabilizers

We can consider a more general invariant than parity: eigenvalues w.r.t. special subsets of the Pauli group.

The n -qubit Pauli group is

Example: Which two-qubit Paulis is this Bell state a $+1$ eigenstate of?

Bit flip code: stabilizers

Consider our logical states:

Which three-qubit Paulis are these states $+1$ eigenstates of?

Stabilizers

Let S be a subgroup of $\mathcal{P}_n$.

Let V_S be a set of states that are $+1$ eigenstates for all $P \in S$.

Then, S is the **stabilizer** of V_S , and V_S is stabilized by S .

Facts about S :

- $-I$ is never in S
- all items of S commute
- choosing S uniquely defines the fixed subspace, V_S

Bit flip code: stabilizers generators

Let's determine a *minimal* representation of the group in terms of its **stabilizer generators**.

Bit flip code: stabilizers and logical operations

Logical Z : need a Pauli that

- stabilizes the logical state but is *not* in S
- *commutes* with everything in S
- has the right action on the subspace

Logical X : need a Pauli that

- stabilizes the logical state but is *not* in S
- *commutes* with everything in S
- has the right action on the subspace
- anticommutes with logical Z

Bit flip code: stabilizer measurement and error detection

We can use this formalism to construct circuits for error detection and recovery: simply *measure the stabilizer generators*.

Bit flip code: stabilizer measurement and error detection

Correctible errors all anticommute with at least one of the generators, so we can detect their presence in the syndrome measurement.

Bit flip code: stabilizers and logical errors

Danger: error that *commutes* with all elements of S , but isn't in S .

Phase flip code: stabilizer formalism

Shor code: stabilizer formalism

Name	Operator
g_1	$ZZIIIIII$
g_2	$IZZIIII$
g_3	$III ZZIII$
g_4	$IIII ZZII$
g_5	$IIII IZZI$
g_6	$IIII I IZZ$
g_7	$XXXXX III$
g_8	$II XXXXXX$
$\tilde{Z}$	$XXXXXXXXX$
$\tilde{X}$	$ZZZZZZZZ$

Screenshot: Nielsen & Chuang, chapter 10.5.6.

The Hamming bound

What is the smallest number of physical qubits that we can use to make a logical qubit, and correct any single-qubit error?

General bound:

- n physical qubits
- k logical qubits
- up to t errors

The smallest code: $[[5, 1, 3]]$

$$|0_L\rangle = \frac{1}{4} \left[|00000\rangle + |10010\rangle + |01001\rangle + |10100\rangle \right. \\ \left. + |01010\rangle - |11011\rangle - |00110\rangle - |11000\rangle \right. \\ \left. - |11101\rangle - |00011\rangle - |11110\rangle - |01111\rangle \right. \\ \left. - |10001\rangle - |01100\rangle - |10111\rangle + |00101\rangle \right]$$

$$|1_L\rangle = \frac{1}{4} \left[|11111\rangle + |01101\rangle + |10110\rangle + |01011\rangle \right. \\ \left. + |10101\rangle - |00100\rangle - |11001\rangle - |00111\rangle \right. \\ \left. - |00010\rangle - |11100\rangle - |00001\rangle - |10000\rangle \right. \\ \left. - |01110\rangle - |10011\rangle - |01000\rangle + |11010\rangle \right]$$

Name	Operator
g_1	$XZZXI$
g_2	$IXZZX$
g_3	$XIXZZ$
g_4	$ZXIXZ$
$\bar{Z}$	$ZZZZZ$
$\bar{X}$	$XXXXX$

Screenshot: Nielsen & Chuang, chapter 10.5.6.

Properties of stabilizer codes

Stabilizer code usually described by notation $[[n, k, d]]$:

- code has $n - k$ stabilizer generators
- d = distance (minimum weight of Paulis that commute with everything in S but aren't in S)
- a distance d code can correct $(d - 1)/2$ errors

Next time

Next class (last class):

- More on stabilizer codes; fault-tolerant quantum computing

Action items:

- ① TA4 due Friday at 23:59
- ② Work on project

Recommended reading:

- From this class: Codebook EC; N&C 10.1-10.5
- For next class: Codebook EC; N&C 10.6